MIL-S-10897D
30 October 1969
SUPERSEDING
MIL-S-10897C
8 March 1967

MILITARY SPECIFICATION

SCOOPS, ICE CREAM, MECHANICAL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope.- This specification covers one type of mechanical ice cream scoop (see 6.1).

1.2 Classification.- The ice cream scoops shall be of the following sizes as specified (see 6.2).

Size 6 - 6 scoops per quart.
Size 8 - 8 scoops per quart.
Size 12 - 12 scoops per quart.
Size 16 - 16 scoops per quart.
Size 30 - 30 scoops per quart.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-665 - Boxes, Fiberboard, Six or Eight Sides.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.

FSC 7330

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

Specification ASTM A167-63 - Corrosion-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip.

Specification ASTM A276-67 - Stainless and Heat-Resisting Steel Bars, Hot-Finished or Cold-Finished.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19106.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, ATTN: Tariff Order Section, 1616 P. Street, N.W., Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Ill., 60606.)

NATIONAL SANITATION FOUNDATION (NSF)

Standard No. C-2-Basic Criteria for the Evaluation of

Special Equipment and/or Devices.

Listing of Food Service Equipment.

(Application for copies should be addressed to the National Sanitation Foundation, P.O. Box 1468, Ann Arbor, Michigan 48106.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article.- When specified (see 6.2), before production is commenced, a sample ice cream scoop covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. Approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Standard product.- The ice cream scoops delivered under this specification shall be the manufacturers' standard commercial product, except for any changes necessary to comply with specification requirements.

3.3 Standard compliance.- The ice cream scoops shall conform to the applicable requirements of NSF Basic Criteria No. C-2.

* 3.3.1 NSF certification.- The supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the ice cream scoop he proposes to furnish under this specification meets the applicable requirements of NSF Basic Criteria C-2 as follows:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF Seal on the finished ice cream scoop shall be submitted prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, or

(2) A certification for the ice cream scoop issued by NSF under their special one-time contract evaluation/certification service shall be submitted prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the ice cream scoops are being procured, indicating that the ice cream scoops have been tested and conform to NSF Basic Criteria C-2 shall be submitted prior to the approval of the preproduction sample, if one is submitted, or prior to commencing production.

* 3.4 Materials.- Materials not definitely specified shall be of the quality normally used by the manufacturer for his ice cream scoop, provided the completed item complies with all provisions of this specification. Commercial tolerances are applicable to all material thicknesses and diameters.

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* 3.4.1 Steel, corrosion-resisting.-

3.4.1.1 Bar.- Corrosion-resisting round steel bar shall conform to the chemical composition of type 302, 303, 304, or 430 of ASTM Specification A276-67.

3.4.1.2 Sheet and strip.- Corrosion-resisting steel sheet and strip shall conform to the chemical composition of type 302, 303, or 304 of ASTM Specification A167-63.

* 3.5 Design and construction.- The ice cream scoops shall be designed as specified herein and as illustrated in figure 1. Each size scoop shall consist of a bowl, bowl knife assembly, handle assembly, and spring-loaded thumb pad for operating the knife assembly.

* 3.5.1 Bowl.- The bowl shall be fabricated from 0.025-inch minimum-thick corrosion-resisting steel specified in 3.4.1.2. The bowl shall be attached to the shank (see 3.5.3) by spot welding, seam welding, or silver soldering. Each size bowl shall have the capacity listed in table I when tested as specified in 4.4.1.

TABLE I

Size	Capacity (fluid ozs.) ± 5 percent	Color code (handle end)
6	5	white
8	4	grey
12	2-2/3	orange
16	2	green
30	1	black

3.5.2 Bowl knife assembly.- The bowl knife assembly shall consist of a shaft, knife, and gear, and shall be either one piece construction or fabricated as separate pieces. When fabricated of separate pieces, the knife in the bowl shall be mechanically locked to a shaft assembly and shall be removable without the use of tools. The shaft shall have a flat or square end and shall fit a matching hole in the gear. The gear shall be securely staked, welded, or soldered to the shaft to prevent longitudinal movement. When pressure is applied to the thumb pad, the gear, shaft, and knife shall rotate against the action of a torsion spring. The knife, thumb pad, gear, and quadrant shall be fabricated from corrosion-resisting steel specified in 3.4.1.2. The shaft shall be fabricated from corrosion-resisting steel specified in 3.4.1.1. The spring shall be fabricated from corrosion-resisting steel wire. The knife shall rotate 180 degrees through the empty bowl and be returned to its original position by the action of the torsion spring

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without binding or malfunctioning, when tested as specified in 4.4.2. The knife shall fit the inside contour of the bowl and shall have a clearance of 0.045 inch maximum at the equator and 0.030 inch maximum clearance at the center of the bowl bottom.

* 3.5.3 Handle assembly.- The handle assembly shall consist of a handle, shank, guard, and color plug as illustrated on figure 1. The handle shall be constructed from black, thermosetting plastic, nylon, or polypropylene, molded or threaded to a shank. The shank shall be fabricated from corrosion-resisting steel specified in 3.4.1.1. The guard shall be fabricated from corrosion-resisting steel specified in 3.4.1.2. The grip of the handle shall be grooved and shall have a colored plastic plug as specified in table I, located in the base of the handle to identify the size of the scoop.

* 3.6 Finish.- All exposed metallic surfaces shall be smooth and shall be free from discoloration, rust, or stains. The molded handle shall be smooth.

* 3.7 Marking.- Each ice cream scoop shall be permanently marked with the manufacturer's name, trade name, or trademark of such known character as to be readily identifiable with the manufacturer. Surfaces opposite stamped markings shall show no signs of penetration or unevenness.

3.8 Workmanship.- Metallic surfaces shall be clean and not contain any rough or sharp edges, burrs, slivers, scratches, rough die marks, gouges, or scale. All surfaces to be welded shall be clean. Welding shall be accomplished in a manner that will prevent pits or flashes. All scale or flux deposits shall be removed from finished welds. The molded surfaces shall be free from blisters, cracks, dirt, and flash, and all gates shall be trimmed flush. The finished ice cream scoop shall not be fractured, buckled, punctured, chipped, scratched, or malformed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.- When a preproduction sample is required, it shall be examined for defects listed in table II and dimensions specified,

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and tested as specified in 4.4.1 and 4.4.2. Failure to pass any test shall be cause for rejection of the preproduction sample. The presence of any defects listed in table II or any dimensions not within specified requirements shall be cause for rejection of the preproduction sample.

4.3 Inspection.- Sampling for inspection shall be made in accordance with provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

* 4.3.2 End product inspection.- The inspection lot shall consist of all the scoops of the same size offered for inspection at one time. The sample unit for this inspection shall be one completely fabricated scoop.

4.3.2.1 Visual examination.- Examination shall be made of the end item for the defects listed in table II. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE II.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish (Metallic and molded surfaces, as applicable)	Not as specified.		X
	Evidence of discoloration, or stains.	X	
	Not smooth on all surfaces.	X	
Construction and workmanship (applicable to all components and assemblies)	Any rough or sharp edges, burrs, slivers, scratches, rough die marks, gouges, pits, rust, or scale.	X	
	Any component missing.	X	
	Any component not securely attached.	X	
	Any component fractured, buckled, punctured, distorted, or malformed.	X	
Bowl	Not spot welded, silver soldered, or seam welded to shank.	X	

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TABLE II.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship (applicable to all components and assemblies) (cont'd)			
Bowl knife assembly	Knife not locked into shaft assembly.		X
	Knife broken.	X	
	Gear loose.		X
	Mechanism binds.	X	
	Torsion spring does not return knife to its original position.	X	
Handle	Not black in color.		X
	Chipped or scratched.		X
	Grooves on handle missing.	X	
	Color plug in handle base not as specified.	X	
Welding	Missing, broken, or burned through.		X
Soldering	Pinholes.		X
Identification marking	Missing, incomplete, or illegible.		X
	Surface opposite marking shows penetration.	X	
	Surface opposite marking shows unevenness.		X

4.3.2.2 Dimensional examination.- Inspection shall be made of the scoop for compliance with dimensions specified herein. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.3.2.3 Testing of the end item.- The end item shall be tested as specified in 4.4.1 and 4.4.2. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

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4.3.3 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be scored in accordance with defects set forth in table III. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

TABLE III.- Examination of preparation for delivery

Examination	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any nonconforming component; component missing or damaged.
Content per container	Number of packages per container is more or less than required.

* 4.3.3.1 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and strapping shall be in accordance with PPP-B-636.

* 4.3.4 Examination of palletization.- An examination shall be made to determine that palletization is in compliance with Section 5 of this specification. Defects shall be scored as specified in table IV. The sample unit for this examination shall be one palletized unit load ready for shipment. The lot shall be the number of palletized unit loads submitted for inspection at one time. The inspection level shall be S-1 with an AQL of 6.5 expressed in terms of defects per hundred units.

TABLE IV.- Examination of palletization

Examination	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

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4.3.5 Code and standards compliance.- Proof of compliance with the requirements of NSF, as specified in 3.3, shall be available to the Government representative.

4.4 Tests.-

4.4.1 Capacity test.- The bowl shall be filled to capacity with water and the volume of the water measured in a graduated beaker to determine compliance with 3.5.1. Inability of the bowl to hold the specified volume of water shall constitute failure of the test.

4.4.2 Operational test.- The thumb pad of the ice cream scoop shall be depressed to its limit and released 25 times. During and after the test, the ice cream scoop shall be inspected for conformance with the requirements specified in 3.5.2. Inability of the knife to rotate 180 degrees and failure to return without binding or malfunctioning shall constitute failure of the test.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Unit package.- Each scoop shall be completely wrapped in anti-tarnish tissue paper. The wrapped scoop shall then be packaged in a snug-fitting folding or metal stayed paperboard box conforming to variety 1, style II, type A or B, class a of PPP-B-566, or style C of PPP-B-665, respectively.

* 5.1.1.2 Intermediate package.- Twelve scoops, unit packaged as specified in 5.1.1.1, shall be packaged in a snug-fitting folding or metal-stayed paperboard box conforming to variety 1, style III or VIII, type B, class a of PPP-B-566, or style A of PPP-B-665, respectively. Closure shall be in accordance with the applicable box specification or appendix thereto.

5.1.2 Level C.- Scoops shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.- Seventy-two scoops, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Closure, waterproofing by means of tape, and reinforcing with flat strapping or tape banding, shall be in accordance with the appendix of PPP-B-636.

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5.2.2 Level B.- Seventy-two scoops, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic, grade 200 of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636.

* 5.2.2.1 When specified (see 6.2), the shipping container shall be V3c, V3s or V4s fabricated and closed in accordance with PPP-B-636.

5.2.3 Level C.- Scoops, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

* 5.3 Palletization.- Unless otherwise specified (see 6.2), scoops packed as specified shall be palletized in accordance with Load Type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The ice cream scoops covered by this specification are intended for dispensing and serving bulk ice cream or shredded, diced, or mashed vegetables.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of scoop required (see 1.2).
(c) First article.- When a preproduction sample is required (see 3.1).
(d) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(e) Whether weather-resistant class fiberboard shipping containers are required for level B shipment (see 5.2.2.1).
(f) When palletization is not required (see 5.3).

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6.3 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 84

Review activities:

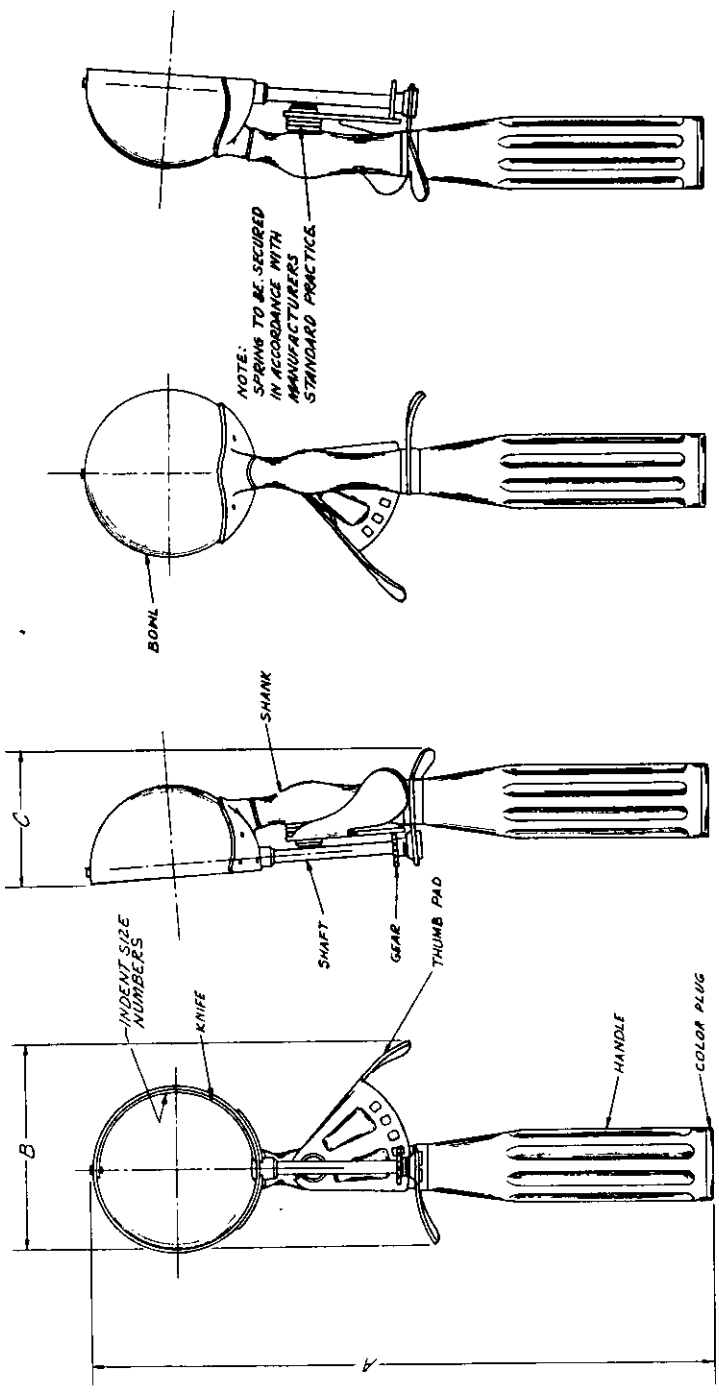
Army - MD
Navy - SA, MC, MS
Air Force - 45

Preparing activity:

Army - GL

Project No. 7330-0276

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DIMENSION IN INCHES

	A		B		C	
	MIN	MAX	MIN	MAX	MIN	MAX
8	9 5/16		2 5/8	3 1/2	1 1/16	2

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FIGURE-1

SCOOP, ICE CREAM, MECHANICAL

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
Scoops, Ice Cream, Mechanical MIL-S-10897D

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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